EXPERIMENT 08
Stochastic Part-of-Speech Tagging

Aim

To demonstrate probabilistic POS tagging by choosing the most likely tag for each word from a small model.

Procedure

1. Create a small dictionary of word-tag probabilities.
2. Choose the tag with the highest probability for each word.
3. Use a default tag for unknown words.
4. Display each word with its selected tag.

Program

```
tag_probabilities = {
    "time": {"NN": 0.8, "VB": 0.2},
    "flies": {"NNS": 0.4, "VBZ": 0.6},
    "like": {"IN": 0.7, "VB": 0.3},
    "birds": {"NNS": 0.9, "VBZ": 0.1}
}

sentence = "time flies like birds".split()
for word in sentence:
    probabilities = tag_probabilities.get(word.lower(), {"NN": 1.0})
    tag = max(probabilities, key=probabilities.get)
    print(word, "->", tag)
```

Output

```
OUTPUT — EXPERIMENT 08
time -> NN
flies -> VBZ
like -> IN
birds -> NNS
```

Result

Thus, a simple stochastic model assigned POS tags using the highest probability.